Future Scope

- Wavelet-based denoising: Improve weak Doppler peak detection under low-SNR conditions.
- ML-Based Peak Detection: Improve atmospheric echo identification in multi-peak spectra.
- Making synthesised radar data more realistic by adding more features
- Distinguish rain echoes from clear-air atmospheric echoes in Doppler spectra to improve the reliability of wind-velocity estimation
- Can be used to fill gaps between observations in the lower atmosphere while the experiment is being conducted at higher altitudes
- Use synthetic data to train reinforcement learning-based algorithms for estimating wind profiles.